EMBEDDING OF THE TAU-FIXED CENTRAL-EXTENSION ALGEBRA D(2N) INTO A(2N)

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Allowed standard prerequisites (editorial).

- PBW theorem, affine root systems, universal central extensions. Original MRY, GNW and HU bibliography entries; author formulations and uses preserved

1. Universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$

In this section, we recall the work of [18], which gives a presentation of the universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$. For a perfect Lie algebra $\mathfrak{g}$, a central extension of $\mathfrak{g}$ is a pair consisting of a Lie algebra $\widetilde{\mathfrak{g}}$ and a surjective homomorphism $\pi\colon \widetilde{\mathfrak{g}} \to \mathfrak{g}$ whose kernel is contained in the center of $\widetilde{\mathfrak{g}}$. The central extension $(\widetilde{\mathfrak{g}}, \pi)$ is called the universal central extension of $\mathfrak{g}$ if for every central extension $(\mathfrak{e}, \psi)$ there exists a unique homomorphism $\phi\colon \widetilde{\mathfrak{g}} \to \mathfrak{e}$ satisfying that $\psi \circ \phi = \pi$.

In [11], Kassel gave an explicit presentation of the universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$. We set $\big(\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big), d\big)$ as the module of differentials of $\mathbb{C}\big[u^{\pm 1}, v\big]$ in the same way as in [18, Section 2]. We take $\{a_i\}$ as a basis of $\mathbb{C}\big[u^{\pm 1}, v\big]$. Let F be a free module of $\mathbb{C}\big[u^{\pm 1}, v\big]$ whose basis is $\big\{\widetilde{d}a_i\big\}$. We regard F as a 2-sided $\mathbb{C}\big[u^{\pm 1}, v\big]$ module by setting $a\big(\widetilde{d}b\big) = \big(\widetilde{d}b\big)a$ for any $a, b \in \mathbb{C}\big[u^{\pm 1}, v\big]$. We set a linear map $\widetilde{d}\colon \mathbb{C}\big[u^{\pm 1}, v\big] \to F$ as $\widetilde{d}\big(\sum_i c_i a_i\big) = \sum_i c_i \widetilde{d}a_i$ and K as a submodule of F generated by $\widetilde{d}(ab) = (\widetilde{d}a)b + a\big(\widetilde{d}b\big)$ for $a, b \in \mathbb{C}\big[u^{\pm 1}, v\big]$. We denote F/K by $\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)$. We also set $d\colon \mathbb{C}\big[u^{\pm 1}, v\big] \to \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)$ as a homomorphism induced by $\widetilde{d}$.

Let $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$ be the Lie algebra whose commutator relations are given by

$$\begin{gathered}
[z_1 \otimes a_1, z_2 \otimes a_2] = [z_1, z_2] \otimes a_1 \cdot a_2 + \kappa(z_1, z_2) a_2 d a_1, \\
\big[\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big], \Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]\big] = 0, \\
\big[\Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big], \Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]\big] = 0.
\end{gathered}$$

We can naturally define a surjective homomorphism

$$\pi\colon\ \mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big] \to \mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big].$$

Since the kernel of π is $\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$, π gives a central extension. Moreover, by [11], $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$ together with π form the universal central extension of $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big]$.

Moody–Rao–Yokonuma [18] gave a new presentation of this universal central extension, which we describe below. Let us take a matrix $\big(a^A_{i,j}\big)_{0 \le i,j \le n-1}$ as

$$a^A_{i,j} = \begin{cases} 2 & \text{if } i = j, \\ -1 & \text{if } i = j \pm 1, \\ -1 & \text{if } (i,j) = (0, n-1), (n-1, 0), \\ 0 & \text{otherwise.} \end{cases}$$

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Then, $\bigl(a^A_{i,j}\bigr)_{0\le i,j\le n-1}$ is the Cartan matrix of an affine Lie algebra $\widehat{\mathfrak{sl}}(n) = \mathfrak{sl}(n) \otimes \mathbb{C}\bigl[t^{\pm 1}\bigr] \oplus \mathbb{C}c$ whose commutator relations are given by

$$[x \otimes t^u, y \otimes t^s] = [x, y] \otimes t^{u+s} + u\delta_{u+s,0} \operatorname{tr}(xy)c \qquad \text{for } x, y \in \mathfrak{sl}(n),$$
$$\bigl[c, \widehat{\mathfrak{sl}}(n)\bigr] = 0,$$

where tr is the trace of $\mathfrak{gl}(n)$.

Definition 1.1. We define $\mathfrak{A}(n)$ as the Lie algebra over $\mathbb{C}$ generated by

$$\bigl\{x^{\pm}_{i,r}, h_{i,r} \mid 0 \le i \le n-1,\ r \in \mathbb{Z}_{\ge 0}\bigr\}$$

subject to the following defining relations:

$$[h_{i,r}, h_{j,s}] = 0, \tag{1.1}$$
$$\bigl[x^{+}_{i,r}, x^{-}_{j,s}\bigr] = \delta_{i,j} h_{i,r+s}, \tag{1.2}$$
$$\bigl[h_{i,0}, x^{\pm}_{j,r}\bigr] = \pm a^A_{i,j} x^{\pm}_{j,r}, \tag{1.3}$$
$$\bigl[h_{i,r+1}, x^{\pm}_{j,s}\bigr] = \bigl[h_{i,r}, x^{\pm}_{j,s+1}\bigr], \tag{1.4}$$
$$\bigl[x^{\pm}_{i,r+1}, x^{\pm}_{j,s}\bigr] = \bigl[x^{\pm}_{i,r}, x^{\pm}_{j,s+1}\bigr], \tag{1.5}$$
$$\sum_{\sigma \in S_{1-a_{i,j}}} \bigl(\operatorname{ad} x^{\pm}_{i,r_{\sigma(1)}}\bigr) \cdots \bigl(\operatorname{ad} x^{\pm}_{i,r_{\sigma(1-a_{i,j})}}\bigr)\bigl(x^{\pm}_{j,s}\bigr) = 0 \qquad \text{if } i \ne j, \tag{1.6}$$

where S_l is a symmetric group of degree l. We also denote the universal enveloping algebra of $\mathfrak{A}(n)$ by $A(n)$.

By using (1.5), we obtain

$$\bigl[x^{\pm}_{i,r}, x^{\pm}_{i,s}\bigr] = \frac{1}{2}\bigl(\bigl[x^{\pm}_{i,r+s}, x^{\pm}_{i,0}\bigr] + \bigl[x^{\pm}_{i,0}, x^{\pm}_{i,r+s}\bigr]\bigr) = 0.$$

Moody–Rao–Yokonuma showed that $\mathfrak{A}(n)$ is isomorphic to the universal central extension of $\mathfrak{sl}(n) \otimes \mathbb{C}\bigl[u^{\pm 1}, v\bigr]$.

Theorem 1.2 ([18, Proposition 3.5])**.** *There exists an isomorphism*

$$\iota\colon\ \mathfrak{A}(n) \to \mathfrak{sl}(n) \otimes \mathbb{C}\bigl[u^{\pm 1}, v\bigr] \oplus \Omega\bigl(\mathbb{C}\bigl[u^{\pm 1}, v\bigr]\bigr)/d\mathbb{C}\bigl[u^{\pm 1}, v\bigr]$$

given by

$$x^{+}_{i,r} = \begin{cases} E_{i,i+1}v^r & \text{if } i \ne 0, \\ E_{n,1}uv^r & \text{if } i = 0, \end{cases} \qquad x^{-}_{i,r} = \begin{cases} E_{i+1,i}v^r & \text{if } i \ne 0, \\ E_{1,n}u^{-1}v^r & \text{if } i = 0, \end{cases}$$

where $E_{i,j}$ is a matrix unit of $\mathfrak{gl}(n)$ whose (p,q) component is $\delta_{p,i}\delta_{q,j}$.

Finally, we consider a presentation of the subalgebra of $A(n)$ generated by $\bigl\{x^{+}_{i,r}\bigr\}$. We take a subspace $\Omega_+ \subset \Omega\bigl(\mathbb{C}\bigl[u^{\pm 1}, v\bigr]\bigr)/d\mathbb{C}\bigl[u^{\pm 1}, v\bigr]$ spanned by $\bigl\{u^{r-1}v^s du \mid r \ge 1,\ s \ge 0\bigr\}$. We set a subalgebra $A(n)_+ \subset A(n)$ generated by $\bigoplus_{s\ge 0}\bigoplus_{\alpha\in\Delta^+} \mathfrak{g}_\alpha v^s$ and Ω_+. Let us take a grading on $A(n)$ by $\deg(h_{i,r}) = \deg\bigl(x^{\pm}_{i,r}\bigr) = r+1$ and denote the graded algebra associated with $A(n)$ by $\operatorname{gr} A(n)$.

Theorem 1.3. *The associative algebra $A(n)_+$ is isomorphic to the associative algebra generated by $\bigl\{\bar{x}^{+}_{i,r} \mid 1 \le i \le n-1,\ r \ge 0\bigr\}$ with the relations:*

$$\bigl[\bar{x}^{+}_{i,r+1}, \bar{x}^{+}_{j,s}\bigr] = \bigl[\bar{x}^{+}_{i,r}, \bar{x}^{+}_{j,s+1}\bigr],$$
$$\bigl[\bar{x}^{+}_{i,r}, \bar{x}^{+}_{j,s}\bigr] = 0 \qquad \text{if } a^A_{i,j} = 0,$$
$$\bigl[\bar{x}^{+}_{i,r}, \bigl[\bar{x}^{+}_{i,s}, \bar{x}^{+}_{j,u}\bigr]\bigr] = 0 \qquad \text{if } a^A_{i,j} = -1.$$

Let $\mathfrak{A}(n)_{\mathrm{pre}}$ be the Lie algebra generated by $\{h_{i,r}, x^{\pm}_{i,r} \mid 1 \le i \le n-1,\, r \ge 0\}$ with the defining relations (1.1)–(1.5). We also define $\mathfrak{A}(n)_{\mathrm{pre},\pm}$ and $\mathfrak{A}(n)_{\mathrm{pre},0}$ as Lie subalgebras generated by $\{x^{\pm}_{i,r}, h_{i,r} \mid 1 \le i \le n-1,\, r \ge 0\}$. Theorem 1.3 follows from the following lemma.

Lemma 1.4.

(1) *The associative algebra $U(\mathfrak{A}(n)_{\mathrm{pre},\pm})$ is isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{\pm}_{i,r}\}$ divided by* (1.5).
(2) *Let $I_{\pm}$ be an ideal of $\mathfrak{A}(n)_{\mathrm{pre}}$ generated by* (1.6). *Then, $I_- \cap A(n)_{\mathrm{pre},+} = \varnothing$.*

Proof. (1) Let V be a quotient algebra of a noncommutative polynomial ring $\mathbb{C}\langle y_{i,d} \mid 0 \le i \le n-1, d \in \mathbb{Z}_{\ge 0}\rangle$ divided by the relations $[y_{i,r+1}, y_{j,s}] - [y_{i,r}, y_{j,s+1}]$. By a direct computation, we obtain a homomorphism

$$\kappa\colon\ \mathfrak{A}(n)_{\mathrm{pre}} \to \mathfrak{gl}(V)$$

by

$$\bigl(\kappa\bigl(x^{-}_{i,d}\bigr)\bigr)(y_{i_1,d_1}\cdots y_{i_r,d_r}) = y_{i,d}y_{i_1,d_1}\cdots y_{i_r,d_r},$$

$$(\kappa(h_{i,d}))(y_{i_1,d_1}\cdots y_{i_r,d_r}) = -\sum_{u=1}^{r} a^{A}_{i,i_u} y_{i_1,d_1}\cdots y_{i_{u-1},d_{u-1}} y_{i_u,d+d_u} y_{i_{u+1},d_{u+1}}\cdots y_{i_r,d_r},$$

$$\bigl(\kappa\bigl(x^{+}_{i,d}\bigr)\bigr)(1) = 0,$$

$$\bigl(\kappa\bigl(x^{+}_{i,d}\bigr)\bigr)(y_{i_1,d_1}\cdots y_{i_r,d_r}) = \delta_{i,i_1}\kappa(h_{i,d+d_1})(y_{i_2,d_2}\cdots y_{i_r,d_r}) + y_{i_1,d_1}\bigl(\kappa\bigl(x^{+}_{i,d}\bigr)\bigr)(y_{i_2,d_2}\cdots y_{i_r,d_r}).$$

The homomorphism κ induces a surjective homomorphism $\Xi\colon \mathfrak{A}(n)_{\mathrm{pre},-} \to V$ by $\Xi(x) = (\kappa(x))1$. By a direct computation, we can also construct a homomorphism

$$\xi\colon\ V \to \mathfrak{A}(n)_{\mathrm{pre},-}, \qquad y_{i,d} \mapsto x^{-}_{i,d}.$$

Since $\xi \circ \Xi$ is an identity map, Ξ is bijective. Then, we find that $U(\mathfrak{A}(n)_{\mathrm{pre},-})$ is isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{-}_{i,r}\}$ divided by (1.5). Since $\mathfrak{A}(n)_{\mathrm{pre},+}$ is isomorphic to $\mathfrak{A}(n)_{\mathrm{pre},-}$, $U(\mathfrak{A}(n)_{\mathrm{pre},+})$ is also isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{+}_{i,r} \mid r \ge 0\}$ divided by (1.5).

(2) It is enough to show that I_- is contained in $\mathfrak{A}(n)_{\mathrm{pre},-}$. This follows from the fact that $\mathrm{ad}\bigl(x^{+}_{i,r}\bigr)$ acts on the left-hand side of (1.6) trivially. □

2. A central extension of $\mathfrak{sl}(2n)\bigl[u^{\pm 1}, v\bigr]^{\tau}$

We give another involution of $\mathfrak{sl}(2n)$. Set $I_n = \{\pm 1, \pm 2, \ldots, \pm n\}$ and take $\{E_{i,j}\}_{i,j\in I_n}$ as the set of matrix units in $\mathfrak{gl}(2n)$. We also define τ as an automorphism of $\mathfrak{gl}(2n)\bigl[u^{\pm 1}, v\bigr]$ by $\tau(E_{i,j}u^r v^s) = -(-1)^s E_{-j,-i}u^r v^s$. By definition of τ, $\mathfrak{sl}(2n)\bigl[u^{\pm 1}, v\bigr]$ is preserved by τ. In the next section, we give a central extension of $\mathfrak{sl}(2n)\bigl[u^{\pm 1}, v\bigr]^{\tau}$.

For $r \ge 0$, we define matrices $\bigl(a^{2r}_{i,j}\bigr)_{0\le i\le n, 0\le j\le n}$ and $\bigl(a^{2r+1}_{i,j}\bigr)_{1\le i\le n-1, 0\le j\le n}$ as follows:

$$a^{2r}_{i,j} = \begin{cases} 2\delta_{i,j} - \delta_{i+1,j} - \delta_{i,j+1} & \text{if } 3 \le i \le n-3, \\ 2\delta_{j,n-2} - \delta_{j,n-3} - \delta_{j,n-1} - \delta_{j,n} & \text{if } i = n-2, \\ 2\delta_{j,n-1} - \delta_{j,n-2} & \text{if } i = n-1, \\ 2\delta_{j,n} - \delta_{j,n-2} & \text{if } i = n, \\ 2\delta_{j,2} - \delta_{j,0} - \delta_{j,1} - \delta_{j,3} & \text{if } i = 2, \\ 2\delta_{j,1} - \delta_{j,2} & \text{if } i = 1, \\ 2\delta_{j,0} - \delta_{j,2} & \text{if } i = 0, \end{cases}$$

$$a_{i,j}^{2r+1} = \begin{cases} a_{i,j}^{2r} & \text{if } 2 \le i \le n-2, \\ 2\delta_{j,n-1} + 2\delta_{j,n} - \delta_{j,n-2} & \text{if } i = n-1, \\ 2\delta_{j,1} + 2\delta_{j,0} - \delta_{j,2} & \text{if } i = 1. \end{cases}$$

We note that $\left(a_{i,j}^{2r}\right)_{i,j=1}^{n}$ is the Cartan matrix of $\widehat{\mathfrak{so}}(2n)$.

Definition 2.1. We define $\mathfrak{D}(2n)$ as the Lie algebra generated by

$$\left\{H_{i,r}, X_{i,r}^{\pm}, H_{n,2r}, H_{0,2r}, X_{-1,2r+1}^{\pm}, X_{n+1,2r+1}^{\pm} \mid 1 \le i \le n-1,\, r \in \mathbb{Z}_{\ge 0}\right\}$$

with the following commutator relations:

$$[H_{i,r}, H_{j,s}] = 0, \tag{2.1}$$

$$\left[H_{i,r}, X_{j,s}^{\pm}\right] = \pm a_{i,j}^{r} X_{j,r+s}^{\pm} \qquad \text{for } 0 \le j \le n, \tag{2.2}$$

(2.3)

$$\left[X_{i,r}^{+}, X_{j,s}^{-}\right] = \begin{cases} \delta_{i,j} H_{i,r+s} & \text{if } 0 \le i, j \le n, \\ & \quad (i,j) \ne (n-1,n), (n,n-1), (0,1), (1,0), (n,n), (0,0), \\ H_{n,r+s} & \text{if } i = j = n \text{ and } r+s \text{ is } \textit{even}, \\ H_{n-1,r+s} & \text{if } i = j = n \text{ and } r+s \text{ is } \textit{odd}, \\ H_{0,r+s} & \text{if } i = j = 0 \text{ and } r+s \text{ is } \textit{even}, \\ H_{1,r+s} & \text{if } i = j = 0 \text{ and } \text{ and } r+s \text{ is } \textit{odd}, \end{cases}$$

(2.4)

$$\left[X_{n-1,r}^{+}, X_{n,s}^{-}\right] = -\delta_{r+s,\mathrm{odd}} X_{n+1,r+s}^{-}, \qquad \left[X_{n,r}^{+}, X_{n-1,s}^{-}\right] = -\delta_{r+s,\mathrm{odd}} X_{n+1,r+s}^{+},$$

$$\left[X_{0,r}^{+}, X_{1,s}^{-}\right] = -\delta_{r+s,\mathrm{odd}} X_{-1,r+s}^{+}, \qquad \left[X_{1,r}^{+}, X_{0,s}^{-}\right] = -\delta_{r+s,\mathrm{odd}} X_{-1,r+s}^{-}, \tag{2.5}$$

$$\left[X_{i,r}^{\pm}, X_{n+1,s}^{+}\right] = 0, \qquad \left[X_{i,r}^{\pm}, X_{n+1,s}^{-}\right] = 0 \qquad \text{if } 1 \le i \le n-1, \tag{2.6}$$

$$\left[X_{i,r}^{\pm}, X_{-1,s}^{+}\right] = 0, \qquad \left[X_{i,r}^{\pm}, X_{-1,s}^{-}\right] = 0 \qquad \text{if } 2 \le i \le n+1, \tag{2.7}$$

$$\left[X_{n-1,r}^{+}, X_{n+1,s}^{+}\right] = 2X_{n,r+s}^{+}, \qquad \left[X_{n-1,r}^{-}, X_{n+1,s}^{+}\right] = 0, \tag{2.8}$$

$$\left[X_{n-1,r}^{+}, X_{n+1,s}^{-}\right] = 0, \qquad \left[X_{n-1,r}^{-}, X_{n+1,s}^{-}\right] = -2X_{n,r+s}^{-}, \tag{2.9}$$

$$\left[X_{n,r}^{+}, X_{n+1,s}^{+}\right] = 0, \qquad \left[X_{n,r}^{-}, X_{n+1,s}^{+}\right] = -2X_{n-1,r+s}^{r+s}, \tag{2.10}$$

$$\left[X_{n,r}^{+}, X_{n+1,s}^{-}\right] = 2X_{n-1,r+s}^{+}, \qquad \left[X_{n,r}^{-}, X_{n+1,s}^{-}\right] = 0, \tag{2.11}$$

$$\left[X_{0,r}^{+}, X_{-1,s}^{+}\right] = 0, \qquad \left[X_{0,r}^{-}, X_{-1,s}^{+}\right] = -2X_{1,r+s}^{-}, \tag{2.12}$$

$$\left[X_{0,r}^{+}, X_{-1,s}^{-}\right] = 2X_{1,r+s}^{+}, \qquad \left[X_{0,r}^{-}, X_{-1,s}^{-}\right] = 0, \tag{2.13}$$

$$\left[X_{1,r}^{+}, X_{-1,s}^{+}\right] = 2X_{0,r+s}^{+}, \qquad \left[X_{1,r}^{-}, X_{-1,s}^{+}\right] = 0, \tag{2.14}$$

$$\left[X_{1,r}^{+}, X_{-1,s}^{-}\right] = 0, \qquad \left[X_{1,r}^{-}, X_{-1,s}^{-}\right] = -2X_{0,r+s}^{-}, \tag{2.15}$$

$$\left[X_{i,r}^{\pm}, X_{i,s}^{\pm}\right] = 0, \tag{2.16}$$

$$\left[X_{i,r+1}^{\pm}, X_{j,s}^{\pm}\right] = \left[X_{i,r}^{\pm}, X_{j,s+1}^{\pm}\right] \quad \text{if } 0 \le i \le j \le n, (i,j) \ne (0,1), (n-1,n), \tag{2.17}$$

(2.18)

$$\left[X_{0,r+1}^{\pm}, X_{1,s}^{\pm}\right] = -\left[X_{0,r}^{\pm}, X_{1,s+1}^{\pm}\right], \qquad \left[X_{n-1,r+1}^{\pm}, X_{n,s}^{\pm}\right] = -\left[X_{n-1,r}^{\pm}, X_{n,s+1}^{\pm}\right],$$

(2.19)

$$\left[X_{i,r}^{\pm}, X_{j,s}^{\pm}\right] = 0 \quad \text{if } a_{i,j}^{0} = 0, 0 \le i \ne j \le n, (i,j) \ne (0,1), (1,0), (n-1,n), (n,n-1),$$

$$\left[X_{i,r}^{\pm}, \left[X_{i,s}^{\pm}, X_{j,u}^{\pm}\right]\right] = 0 \qquad \text{if } a_{i,j}^{0} = -1, \tag{2.20}$$

$$\left[X_{n-1,0}^{\pm}, \left[X_{n-1,0}^{\pm}, X_{n,2r+1}^{\pm}\right]\right] = 0, \tag{2.21}$$

$$\left[X_{n,0}^{\pm}, \left[X_{n,0}^{\pm}, X_{n-1,2r+1}^{\pm}\right]\right] = 0, \tag{2.22}$$

$$\left[X_{1,0}^{\pm}, \left[X_{1,0}^{\pm}, X_{0,2r+1}^{\pm}\right]\right] = 0, \tag{2.23}$$

$$\left[X_{0,0}^{\pm}, \left[X_{0,0}^{\pm}, X_{1,2r+1}^{\pm}\right]\right] = 0, \tag{2.24}$$

$$\big[X^+_{1,1},\big[X^-_{1,0},(E_{1,1}-E_{2,2})t^u\big]\big]-\big[X^+_{1,0},\big[X^-_{1,1},(E_{1,1}-E_{2,2})t^u\big]\big]=0. \tag{2.25}$$

We also denote the universal enveloping algebra of $\mathfrak{D}(2n)$ by $D(2n)$.

Set $f^{r,s}_{i,j}=(E_{i,j}-(-1)^sE_{-j,-i})u^rv^s\in\mathfrak{sl}(2n)\big[u^{\pm1},v\big]$. Then, $\mathfrak{sl}(2n)\big[u^{\pm1},v\big]^\tau$ is spanned by $\big\{f^{r,s}_{i,j}\mid i,j\in I_n,\,r\in\mathbb{Z},\,s\in\mathbb{Z}_{\geq0}\big\}$. By a direct computation, we can construct a homomorphism $\pi\colon\mathfrak{D}(2n)\to\mathfrak{sl}(2n)\big[u^{\pm1},v\big]^\tau$ determined by

$$H_{i,r}=\begin{cases}f^r_{i,i}v^r-f^r_{i+1,i+1}v^r & \text{if } 1\leq i\leq n-1,\\ f^r_{n-1,n-1}v^r+f^r_{n,n}v^r & \text{if } i=n \text{ and } r \text{ is even},\\ -\big(f^r_{1,1}v^r+f^r_{2,2}v^r\big) & \text{if } i=0 \text{ and } r \text{ is even},\end{cases}$$

$$X^+_{i,r}=\begin{cases}f^r_{i,i+1}v^r & \text{if } 1\leq i\leq n-1,\\ f^r_{n-1,-n}v^r & \text{if } i=n,\\ (-1)^rf^r_{-2,1}uv^r & \text{if } i=0,\\ f^r_{n,-n}v^r & \text{if } i=n+1 \text{ and } r \text{ is odd},\\ f^r_{-1,1}uv^r & \text{if } i=-1 \text{ and } r \text{ is odd},\end{cases}$$

$$X^-_{i,r}=\begin{cases}f^r_{i+1,i}v^r & \text{if } 1\leq i\leq n-1,\\ f^r_{-n,n-1}v^r & \text{if } i=n,\\ (-1)^rf^r_{1,-2}u^{-1}v^r & \text{if } i=0,\\ f^r_{-n,n}v^r & \text{if } i=n+1 \text{ and } r \text{ is odd},\\ f^r_{1,-1}u^{-1}v^r & \text{if } i=-1 \text{ and } r \text{ is odd}.\end{cases}$$

Next, we show that $\mathfrak{D}(2n)$ is a central extension of $\mathfrak{sl}(2n)\big[u^{\pm1},v\big]^\tau$.

Let $\{\alpha_i\mid0\leq i\leq n\}$ be the set of simple roots of $\widehat{\mathfrak{so}}(2n)$, let W be the Weyl group of $\widehat{\mathfrak{so}}(2n)$, and let Δ be the set of roots of $\widehat{\mathfrak{so}}(2n)$. We also denote

$$Q=\sum_{i=0}^n\mathbb{Z}\alpha_i,Q_+=\sum_{i=0}^n\mathbb{Z}_{\geq0}\alpha_i\setminus\{0\},\qquad Q_-=-Q_+\qquad\text{and}$$

$$\delta=\alpha_0+\alpha_1+\alpha_{n-1}+\alpha_n+\sum_{j=2}^{n-2}\alpha_j.$$

We also take a subset of Q as follows:

$$\Delta^{\mathrm{ex}}=\Delta\cup\Bigg\{\pm\left(2\sum_{j=i}^{n-2}\alpha_j+\alpha_{n-1}+\alpha_n\right)+s\delta,$$

$$\pm(\alpha_{n-1}-\alpha_n)+s\delta,\pm(\alpha_0-\alpha_1)+s\delta\mid1\leq i\leq n-2,\,s\in\mathbb{Z}\Bigg\}.$$

We define a degree of $\mathfrak{D}(2n)$ by $Q\times\mathbb{Z}_{\geq0}$ by assigning generators as follows:

$$\deg(H_{i,r})=(0,r),\qquad\deg(X^\pm_{i,r})=\begin{cases}(\pm\alpha_i,r) & \text{if } 0\leq i\leq n,\\ (\pm(\alpha_n-\alpha_{n-1}),r) & \text{if } i=n+1,\\ (\pm(\alpha_0-\alpha_1),r) & \text{if } i=0.\end{cases}$$

We denote the space of elements of degree (α,k) in $\mathfrak{D}(2n)$ by $\mathfrak{D}(2n)^k_\alpha$. The following lemma can be proved in the same way as [18, Lemma 3.1].

Lemma 2.2.

(1) *Let $\mathfrak{s}_\pm$ be the subspace of $\mathfrak{D}(2n)$ spanned by $\big\{\prod_{i=1}^v\mathrm{ad}\big(X^\pm_{u_i,r_i}\big)X^\pm_{u_{v+1},r_{v+1}}\big\}$ and $\mathfrak{h}$ as the space spanned by $\{H_{i,r}\}$. Then, $\mathfrak{D}(2n)$ can be decomposed to*

$$\mathfrak{s}^+\oplus\mathfrak{h}\oplus\mathfrak{s}^-\oplus\bigoplus_{r\in\mathbb{Z}_{\geq0}}\mathbb{C}X^+_{n+1,2r+1}\oplus\bigoplus_{r\in\mathbb{Z}_{\geq0}}\mathbb{C}X^-_{n+1,2r+1}$$

$$\oplus \bigoplus_{r\in\mathbb{Z}_{\geq 0}} \mathbb{C}X^+_{-1,2r+1} \oplus \bigoplus_{r\in\mathbb{Z}_{\geq 0}} \mathbb{C}X^-_{-1,2r+1}.$$

(2) *If* $\alpha \in Q_\pm$, $\mathfrak{D}(2n)^k_\alpha$ *is spanned by*

$$\left\{ \prod_{i=1}^{v} \mathrm{ad}\big(X^\pm_{u_i,r_i}\big) X^\pm_{u_{v+1},r_{v+1}} \;\middle|\; \sum_{i=1}^{v+1} \alpha_{u_i} = \alpha,\ \sum_{i=1}^{v+1} r_i = k \right\}.$$

For $0 \leq i \leq n$, we set

$$s_i \in W \qquad \text{as } s_i(\alpha_j) = \begin{cases} -\alpha_i & \text{if } i = j, \\ \alpha_i + \alpha_j & \text{if } a^0_{i,j} = -1, \\ \alpha_j & \text{otherwise.} \end{cases}$$

Lemma 2.3. *By the action of the Weyl group,* Δ^{ex} *is generated by* $\{\alpha_j \mid 0 \leq j \leq n\}$ *and* $\pm(\alpha_{n-1} - \alpha_n)$, $\pm(\alpha_0 - \alpha_1)$.

Proof. Any root of Δ can be derived from $\{\alpha_j \mid 0 \leq j \leq n\}$ by the Weyl group. Thus, it is enough to show that $\pm\big(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n\big) + s\delta$ and $\pm(\alpha_{n-1} - \alpha_n) + s\delta$ can be derived from $\pm(\alpha_{n-1} - \alpha_n)$, $\pm(\alpha_0 - \alpha_1)$.

First, we consider the case where s is even. By the definition of s_i, we obtain the following relations:

$$s_n(\alpha_{n-1} - \alpha_n + s\delta) = \alpha_{n-1} + \alpha_n + s\delta,$$
$$s_{n-1}(\alpha_{n-1} + \alpha_n + s\delta) = -\alpha_{n-1} + \alpha_n + s\delta,$$
$$s_{i-1}\left(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n + s\delta\right) = 2\sum_{j=i-1}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n + s\delta,$$
$$s_0\left(2\sum_{j=1}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n + s\delta\right) = -\alpha_0 - \alpha_1 + (s+1)\delta,$$
$$s_0(-\alpha_0 - \alpha_1 + (s+1)\delta) = \alpha_0 - \alpha_1 + (s+1)\delta,$$
$$s_1(\alpha_0 - \alpha_1 + s\delta) = \alpha_0 + \alpha_1 + s\delta = -2\sum_{j=1}^{n-2} \alpha_j - \alpha_{n-1} - \alpha_n + (s+1)\delta,$$
$$s_i\left(-2\sum_{j=i}^{n-2} \alpha_j - \alpha_{n-1} - \alpha_n + (s+1)\delta\right) = -2\sum_{j=i+1}^{n-2} \alpha_j - \alpha_{n-1} - \alpha_n + (s+1)\delta,$$
$$s_{n-1}(-\alpha_{n-1} - \alpha_n + (s+1)\delta) = \alpha_{n-1} - \alpha_n + (s+1)\delta.$$

Then, by these relations, $\big(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n\big) + s\delta$, $-\big(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n\big) + (s+1)\delta$, $\pm(\alpha_{n-1} - \alpha_n) + s\delta$ and $\pm(\alpha_0 - \alpha_1) + s\delta$ can be derived from $\alpha_{n-1} - \alpha_n$ and $\alpha_0 - \alpha_1$ for $s \geq 0$. Similarly to the positive case, $\big(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n\big) - (s+1)\delta$, $-\big(2\sum_{j=i}^{n-2} \alpha_j + \alpha_{n-1} + \alpha_n\big) - s\delta$, $\pm(\alpha_{n-1} - \alpha_n) - (s+1)\delta$ and $\pm(\alpha_0 - \alpha_1) - (s+1)\delta$ can be derived from $-(\alpha_{n-1} - \alpha_n)$ and $-(\alpha_0 - \alpha_1)$ for $s \geq 0$. □

By the relations (2.19)–(2.24), we can define an automorphism of $D(2n)$ by

$$\tau_i = \exp\big(\mathrm{ad}\big(X^+_{i,0}\big)\big) \exp\big(-\mathrm{ad}\big(X^-_{i,0}\big)\big) \exp\big(\mathrm{ad}\big(X^+_{i,0}\big)\big).$$

By a direct computation, we obtain the following lemma.

Lemma 2.4. *The following relations hold for* $1 \leq i \leq n-2$*:*

$$\tau_i(X^+_{j,r}) = \begin{cases} -X^-_{i,r} & \text{if } i = j, \\ \big[X^+_{i,0}, X^+_{j,r}\big] & \text{if } a^0_{i,j} = -1, \\ (-1)^r X^+_{j,r} & \text{if } (i,j) = (0,1), (1,0), (n-1,n), (n,n-1), \\ X^+_{i,r} & \text{otherwise,} \end{cases}$$

$$\tau_i(X^-_{j,r}) = \begin{cases} -X^+_{i,r} & \text{if } i = j, \\ -\big[X^-_{i,0}, X^-_{j,r}\big] & \text{if } a^0_{i,j} = -1, \\ (-1)^r X^-_{j,r} & \text{if } (i,j) \neq (0,1),(1,0),(n-1,n),(n,n-1), \\ X^+_{j,r} & \text{otherwise.} \end{cases}$$

Proof. We only consider the case where $(i,j) = (n-1,n)$ when r is odd. The other cases can be proved in a similar way. First, we prove the $+$ case. By (2.21), we have

$$\exp\big(\operatorname{ad}\big(X^+_{n-1,0}\big)\big)X^+_{n,2r+1} = X^+_{n,2r+1} + \big[X^+_{n-1,0}, X^+_{n,2r+1}\big] = X^+_{n,2r+1} + \big[X^+_{n-1,0}, X^+_{n,2r+1}\big].$$

By (2.4), (2.2) and (2.8), we obtain

$$\begin{aligned} &-\operatorname{ad}\big(X^-_{n-1,0}\big)\big(X^+_{n,2r+1} + \big[X^+_{n-1,0}, X^+_{n,2r+1}\big]\big) \\ &\quad = -\big[X^-_{n-1,0}, X^+_{n,2r+1}\big] - \big[X^-_{n-1,0}, \big[X^+_{n-1,0}, X^+_{n,2r+1}\big]\big] \\ &\quad = -X^+_{n+1,2r+1} + \big[h_{n-1,0}, X^+_{n,2r+1}\big] - \big[X^+_{n-1,0}, \big[X^-_{n-1,0}, X^+_{n,2r+1}\big]\big] \\ &\quad = -X^+_{n+1,2r+1} - \big[X^+_{n-1,0}, X^+_{n+1,2r+1}\big] = X^+_{n+1,2r+1} - 2X^+_{n,2r+1}. \end{aligned}$$

By (2.8) and (2.4), we have

$$-\frac{1}{2}\operatorname{ad}\big(X^-_{n-1,0}\big)\big(-X^+_{n+1,2r+1} - 2X^+_{n,2r+1}\big) = 0 + X^+_{n+1,2r+1}.$$

Thus, we obtain

$$\exp\big(-\operatorname{ad}\big(X^-_{n-1,0}\big)\big)\exp\big(\operatorname{ad}\big(X^+_{n-1,0}\big)\big)X^+_{n,2r+1} = -X^+_{n,2r+1} + \big[X^+_{n-1,0}, X^+_{n,2r+1}\big].$$

By (2.21), we obtain

$$\begin{aligned} &\operatorname{ad}\big(X^+_{n-1,0}\big)\big(-X^+_{n,2r+1} + \big[X^+_{n-1,0}, X^+_{n,2r+1}\big]\big) \\ &\quad = -\big[X^+_{n-1,0}, X^+_{n,2r+1}\big] + \big[X^+_{n-1,0}, \big[X^+_{n-1,0}, X^+_{n,2r+1}\big]\big] = -\big[X^+_{n-1,0}, X^+_{n,2r+1}\big] + 0, \\ &\operatorname{ad}\big(X^+_{n-1,0}\big)\big(-[X^+_{n-1,0}, X^+_{n,2r+1}]\big) = 0. \end{aligned}$$

Thus, we have obtained $\tau_{n-1}\big(X^+_{n,2r+1}\big) = -X^+_{n,2r+1}$.

Next, we show the $-$ case. By (2.4) and (2.9), we have

$$\begin{aligned} \exp\big(\operatorname{ad}\big(X^+_{n-1,0}\big)\big)X^-_{n,2r+1} &= X^-_{n,2r+1} + \big[X^+_{n-1,0}, X^-_{n,2r+1}\big] + \frac{1}{2}\big[X^+_{n-1,0}, \big[X^+_{n-1,0}, X^-_{n,2r+1}\big]\big] \\ &= X^-_{n,2r+1} - X^-_{n+1,2r+1} + 0. \end{aligned}$$

By (2.9), we obtain

$$\begin{aligned} \big(-\operatorname{ad}\big(X^-_{n-1,0}\big)\big)\big(X^-_{n,2r+1} - X^-_{n+1,2r+1}\big) &= -\big[X^-_{n-1,0}, X^-_{n,2r+1}\big] + \big[X^-_{n-1,0}, X^-_{n+1,2r+1}\big] \\ &= -\big[X^-_{n-1,0}, X^-_{n,2r+1}\big] - 2X^-_{n,2r+1}. \end{aligned}$$

By (2.21), we have

$$-\frac{1}{2}\operatorname{ad}\big(X^-_{n-1,0}\big)\big(-\big[X^-_{n-1,0}, X^-_{n,2r+1}\big] - 2X^-_{n,2r+1}\big) = 0 + \big[X^-_{n-1,0}, X^-_{n,2r+1}\big].$$

Then, we obtain

$$\exp\big(-\operatorname{ad}\big(X^-_{n-1,0}\big)\big)\exp\big(\operatorname{ad}\big(X^+_{n-1,0}\big)\big)X^-_{n,2r+1} = -X^-_{n,2r+1} - X^-_{n+1,2r+1}.$$

By (2.4) and (2.9), we obtain

$$\begin{aligned} &\operatorname{ad}\big(X^+_{n-1,0}\big)\big(-X^-_{n,2r+1} - X^-_{n+1,2r+1}\big) = X^-_{n+1,2r+1} - 0, \\ &\frac{1}{2}\big(\operatorname{ad}\big(X^+_{n-1,0}\big)\big)\big(X^-_{n+1,2r+1}\big) = 0. \end{aligned}$$

Thus, we have obtained $\tau_{n-1}\big(X^-_{n,2r+1}\big) = -X^-_{n,2r+1}$. □

Corollary 2.5. *The following relations hold:*

$$\tau_i(f^{r,s}_{j,k}) = \begin{cases} -f^{r,s}_{k,j} & \text{if } (j,k) = (i,i+1),(i+1,i), \\ (-1)^r f^{r,s}_{j,k} & \text{if } (j,k) = (i,-(i+1)),(-(i+1),i),(-(i+1),i),(-i,(i+1)), \\ -f^{r,s}_{j,k} & \text{if } j = i, k \neq \pm(i+1), \\ -f^{r,s}_{-i-1,k} & \text{if } j = -i, k \neq (i+1), \\ f^{r,s}_{i,k} & \text{if } j = i+1, k \neq \pm i, \\ f^{r,s}_{-i,k} & \text{if } j = -(i+1) \neq \pm i, \\ -f^{r,s}_{j,i+1} & \text{if } k = i, j \neq \pm(i+1), \\ -f^{r,s}_{j,-i-1} & \text{if } k = -i, j \neq (i+1), \\ f^{r,s}_{j,i} & \text{if } k = i+1, j \neq i, \\ f^{r,s}_{j,-i} & \text{if } k = -(i+1), j \neq i, \\ f^{r,s}_{j,k} & \text{otherwise}, \end{cases}$$

$$\tau_n(f^{r,s}_{j,k}) = \begin{cases} (-1)^r f^{r,s}_{j,k} & \text{if } (j,k) = (n-1,n),(n,n-1), \\ -f^{r,s}_{k,j} & \text{if } (j,k) = (n-1,-n),(n,-(n-1)),(-(n-1),n), \\ -f^{r,s}_{-n,k} & \text{if } j = n-1, k \neq \pm n, \\ -f^{r,s}_{n,k} & \text{if } j = -(n-1), k \neq n, \\ f_{-(n-1),k} & \text{if } j = n, k \neq \pm n-1, \\ f^{r,s}_{(n-1),k} & \text{if } j = -n \neq \pm n-1, \\ -f^{r,s}_{j,-n} & \text{if } k = n-1, j \neq \pm n, \\ -f^{r,s}_{j,n} & \text{if } k = -(n-1), j \neq n, \\ f^{r,s}_{j,-(n-1)} & \text{if } k = n, j \neq n-1, \\ f^{r,s}_{j,(n-1)} & \text{if } k = -n, j \neq n-1, \\ f^{r,s}_{j,k} & \text{otherwise}, \end{cases}$$

$$\tau_0(f^{r,s}_{j,k}) = \begin{cases} (-1)^r f^{r,s}_{j,k} & \text{if } (j,k) = (1,2),(j,k) = (2,1), \\ -f^{r-2,s}_{k,j} & \text{if } (j,k) = (1,-2),(1,-2), \\ -f^{r+2,s}_{k,j} & \text{if } (j,k) = (-2,1),(-1,2), \\ f^{r+2,s}_{-2,j} & \text{if } j = 1, k \neq \pm 2, \\ f^{r-2,s}_{2,k} & \text{if } j = -1, k \neq 2, \\ -f^{r+1,s}_{-1,k} & \text{if } j = 2, k \neq \pm 1, \\ -f^{r-1,s}_{1,k} & \text{if } j = -2 \neq 1, \\ f^{r-1,s}_{j,-2} & \text{if } k = 1, j \neq \pm 2, \\ f^{r+1,s}_{j,2} & \text{if } k = -1, j \neq 2, \\ -f^{r-1,s}_{j,-1} & \text{if } k = 2, j \neq 1, \\ -f^{r+1,s}_{j,1} & \text{if } k = -2, j \neq 1, \\ f^{r,s}_{j,k} & \text{otherwise}, \end{cases}$$

for $j \neq \pm k$ and either j or k is positive.

By Lemma 2.4, we find that $\tau_i\big(\mathfrak{D}(2n)^k_\alpha\big)$ is contained in $\mathfrak{D}(2n)^k_{s_i(\alpha)}$ In the same way as [18, Proposition 3.2], we obtain the following theorem.

Theorem 2.6. *We find that*

$$\begin{cases} \dim \mathfrak{D}(2n)^k_\alpha = 1 & \text{if } \alpha \in \Delta^{\mathrm{ex}} \setminus \{k\delta \mid k \in \mathbb{Z} \setminus \{0\}\}, \\ \dim \mathfrak{D}(2n)^k_\alpha = 0 & \text{if } \alpha \notin \Delta^{\mathrm{ex}}. \end{cases}$$

By Theorem 2.6, the kernel of π is contained in $\bigoplus_{r\in\mathbb{Z}}\mathfrak{D}(2n)^k_{r\delta}$. Since $\big[X^{\pm}_{i,0},\mathfrak{D}(2n)^k_{r\delta}\big]\cap\mathfrak{D}(2n)^k_{r\delta}$ is empty by Lemma 2.2, the kernel of π is contained in the center of $D(2n)$. Thus, we find that $\mathfrak{D}(2n)$ is a central extension of $\mathfrak{sl}(2n)\big[u^{\pm1},v\big]^{\tau}$.

3. Embedding of $D(2n)$ into $A(2n)$

In this section, we show that the associative algebra $\mathfrak{D}(2n)$ is a subalgebra of the associative algebra $A(2n)$. By a direct computation, we can construct a homomorphism $\phi\colon\mathfrak{D}(2n)\to A(2n)$ defined by

$$H_{i,r}=\begin{cases}f^r_{i,i}v^r-f^r_{i+1,i+1}v^r & \text{if } 1\le i\le n-1,\\ f^r_{n-1,n-1}v^r+f^r_{n,n}v^r & \text{if } i=n \text{ and } r \text{ is even},\\ -\big(f^r_{1,1}v^r+f^r_{2,2}v^r\big)+2u^{-1}v^r du & \text{if } i=0 \text{ and } r \text{ is even},\end{cases}$$

$$X^+_{i,r}=\begin{cases}f^r_{i,i+1}v^r & \text{if } 1\le i\le n-1,\\ f^r_{n-1,-n}v^r & \text{if } i=n,\\ (-1)^r f^r_{-2,1}uv^r & \text{if } i=0,\\ f^r_{n,-n}v^r & \text{if } i=n+1 \text{ and } r \text{ is odd},\\ f^r_{-1,1}uv^r & \text{if } i=-1 \text{ and } r \text{ is odd},\end{cases}$$

$$X^-_{i,r}=\begin{cases}f^r_{i+1,i}v^r & \text{if } 1\le i\le n-1,\\ f^r_{-n,n-1}v^r & \text{if } i=n,\\ (-1)^r f^r_{1,-2}u^{-1}v^r & \text{if } i=0,\\ f^r_{-n,n}v^r & \text{if } i=n+1 \text{ and } r \text{ is odd},\\ f^r_{1,-1}u^{-1}v^r & \text{if } i=-1 \text{ and } r \text{ is odd}.\end{cases}$$

Theorem 3.1. *The homomorphism ϕ is injective.*

This section is devoted to the proof of Theorem 3.1. By Theorem 2.6, we can define $\widetilde{f}^{r,s}_{i,j}\in\mathfrak{D}(2n)$ for $i\ne j$ in the same way as $f^{r,s}_{i,j}\in\mathfrak{sl}(2n)\big[u^{\pm1},v\big]^{\tau}$. By Theorem 2.6, we obtain the relations

$$\big[H_{i,x},\widetilde{f}^{r,s}_{p,q}\big]$$
$$=\begin{cases}(\delta_{i,p}-\delta_{i+1,p}-\delta_{i,q}+\delta_{i+1,q}-(-1)^x(\delta_{-i,p}-\delta_{-i,q}-\delta_{-i-1,p}+\delta_{-i-1,q}))\widetilde{f}^{r,s+x}_{p,q}\\ \quad\text{if } 1\le i\le n-1,\\ (\delta_{n-1,p}+\delta_{n,p}-\delta_{n-1,q}-\delta_{n,q}-(-1)^x(\delta_{-(n-1),p}-\delta_{-(n-1),q}+\delta_{-n,p}-\delta_{-n,q}))\widetilde{f}^{r,s+x}_{p,q}\\ \quad\text{if } i=n,\\ -(\delta_{1,p}+\delta_{2,p}-\delta_{1,q}-\delta_{2,q}-(-1)^x(\delta_{-1,p}-\delta_{-1,q}+\delta_{-2,p}-\delta_{-2,q}))\widetilde{f}^{r,s+x}_{p,q}\\ \quad\text{if } i=0,\end{cases}$$

$$\big[\widetilde{f}^{r_1,s_1}_{i,j},\widetilde{f}^{r_2,s_2}_{p,q}\big]=\delta_{j,p}\widetilde{f}^{r_1+r_2,s_1+s_2}_{i,q}-\delta_{i,q}\widetilde{f}^{r_1+r_2,s_1+s_2}_{p,j}$$
$$-(-1)^{s_1}\delta_{i,-p}\widetilde{f}^{r_1+r_2,s_1+s_2}_{-j,q}+(-1)^{s_1}\delta_{j,-q}\widetilde{f}^{r_1+r_2,s_1+s_2}_{p,-i}\qquad\text{if } (i,j)\ne(q,p),(-p,-q).$$

By Theorem 2.6, we find that the kernel of Φ is contained in $\bigoplus_{r\ne0,k\in\mathbb{Z}}\mathfrak{D}(2n)^k_{r\delta}$.

Lemma 3.2. *For $r\ge0$, the subspace $\mathfrak{D}(2n)^k_{r\delta}$ is spanned by*

$$a(r,k)=\big\{\big[X^+_{i,0},\widetilde{f}^{r,k}_{i+1,i}\big]\mid1\le i\le n-1\big\}\cup\big\{\big[X^+_{n,0},\widetilde{f}^{r,k}_{-n,n-1}\big]\mid1\le i\le n-1\big\}$$
$$\cup\big\{\big[X^+_{0,s},\widetilde{f}^{r,k}_{1,-2}\big]\mid s=0,1\big\}\cup\big\{\big[X^+_{1,1},\widetilde{f}^{r,k-1}_{2,1}\big]-\big[X^+_{1,0},\widetilde{f}^{r,k}_{2,1}\big]\big\}.$$

Proof. By Theorem 2.6, we find that $\mathfrak{D}(2n)^k_{r\delta}$ is spanned by the set

$$\big\{\big[X^+_{i,s},\widetilde{f}^{r,v}_{i+1,i}\big]\mid1\le i\le n-1,s+v=k\big\}\cup\big\{\big[X^+_{n,s},\widetilde{f}^{r,v}_{-n,n-1}\big]\mid s+v=k\big\}$$
$$\cup\big\{\big[X^+_{0,s},\widetilde{f}^{r-1,v}_{1,-2}\big]\mid s+v=k\big\}.$$

Since $\mathfrak{D}(2n)$ is a central extension of $\mathfrak{sl}(2n)\big[u^{\pm 1}, v\big]^{\tau}$, we find that

$$\begin{gathered}\big[X^+_{i,s}, \widetilde{f}^{r,v}_{i+1,i}\big] - \big[X^+_{i,0}, \widetilde{f}^{r,s+v}_{i+1,i}\big],\\ \big[X^+_{n,s}, \widetilde{f}^{r,v}_{-n,n-1}\big] - \big[X^+_{n,0}, \widetilde{f}^{r,s+v}_{-n,n-1}\big],\\ \big[X^+_{0,s}, \widetilde{f}^{r-1,v}_{1,-2}\big] - \big[X^+_{0,0}, \widetilde{f}^{r-1,s+v}_{1,-2}\big]\end{gathered}$$

are central elements of $\mathfrak{D}(2n)$. Thus, we obtain

$$\begin{aligned}0 &= \big[H_{i,1}, \big[X^+_{i,s}, \widetilde{f}^{r,v}_{i+1,i}\big]\big] - \big[H_{i,1}, \big[X^+_{i,0}, \widetilde{f}^{r,s+v}_{i+1,i}\big]\big] = 2\big[X^+_{i,s+1}, \widetilde{f}^{r,v}_{i+1,i}\big] - 2\big[X^+_{i,s}, \widetilde{f}^{r,v+1}_{i+1,i}\big]\\ &\quad - 2\big[X^+_{i,1}, \widetilde{f}^{r,s+v}_{i+1,i}\big] + 2\big[X^+_{i,0}, \widetilde{f}^{r,s+v+1}_{i+1,i}\big] \qquad \text{for } 1 \le i \le n-1,\\ 0 &= \big[H_{n-1,1}, \big[X^+_{n,s}, \widetilde{f}^{r,v}_{-n,n-1}\big]\big] - \big[H_{n-1,1}, \big[X^+_{n,0}, \widetilde{f}^{r,s+v}_{-n,n-1}\big]\big]\\ &= 2\big[X^+_{n,s+1}, \widetilde{f}^{r,v}_{-n,n-1}\big] - 2\big[X^+_{n,s}, \widetilde{f}^{r,v+1}_{-n,n-1}\big] - 2\big[X^+_{n,1}, \widetilde{f}^{r,s+v}_{-n,n-1}\big] + 2\big[X^+_{n,0}, \widetilde{f}^{r,s+v+1}_{-n,n-1}\big],\\ 0 &= \big[H_{1,1}, \big[X^+_{0,s}, \widetilde{f}^{r-1,v}_{1,-2}\big]\big] - \big[H_{1,1}, \big[X^+_{0,0}, \widetilde{f}^{r-1,s+v}_{1,-2}\big]\big]\\ &= 2\big[X^+_{0,s+1}, \widetilde{f}^{r-1,v}_{1,-2}\big] - 2\big[X^+_{0,s}, \widetilde{f}^{r-1,v+1}_{1,-2}\big] - 2\big[X^+_{0,1}, \widetilde{f}^{r-1,s+v}_{1,-2}\big] + 2\big[X^+_{0,0}, \widetilde{f}^{r-1,s+v+1}_{1,-2}\big].\end{aligned} \tag{3.1}$$

Thus, $\mathfrak{A}(2n)^k_{r\delta}$ is spanned by

$$\begin{aligned}&\big\{\big[X^+_{i,s}, \widetilde{f}^{r,k-s}_{i+1,i}\big] \mid 1 \le i \le n-1,\ s = 0, 1\big\}\\ &\qquad \cup \big\{\big[X^+_{n,s}, \widetilde{f}^{r,k-s}_{-n,n-1}\big] \mid 1 \le i \le n-1, s = 0, 1\big\} \cup \big\{\big[X^+_{0,s}, \widetilde{f}^{r-1,k-s}_{1,-2}\big] \mid s = 0, 1\big\}.\end{aligned}$$

By Corollary 2.5 and a direct computation, we find that, by composing τ_i, we can construct $\widetilde{\tau}_1$, $\widetilde{\tau}_2$ satisfying that

$$\begin{aligned}\widetilde{\tau}_1\big(X^+_{i,k}\big) &= X^+_{1,k}, \qquad \widetilde{\tau}_1\big(\widetilde{f}^{r,k}_{i,i+1}\big) = \widetilde{f}^{r,k}_{2,1},\\ \widetilde{\tau}_2\big(X^+_{0,k}\big) &= X^+_{1,k}, \qquad \widetilde{\tau}_2\big(\widetilde{f}^{r-1,k}_{1,-2}\big) = \widetilde{f}^{r,k}_{2,1},\\ \widetilde{\tau}_3\big(X^+_{n,k}\big) &= X^+_{1,k}, \qquad \widetilde{\tau}_3\big(\widetilde{f}^{r,k}_{-n,n-1}\big) = \widetilde{f}^{r,k}_{2,1}\end{aligned}$$

for $1 \le i \le n$. Then, we obtain

$$\begin{aligned}\big[X^+_{1,1}, \widetilde{f}^{r,k-1}_{2,1}\big] - \big[X^+_{1,0}, \widetilde{f}^{r,k}_{2,1}\big] &= \big[\widetilde{\tau}_1\big(X^+_{i,1}\big), \widetilde{\tau}_1\big(\widetilde{f}^{r,k-1}_{i+1,i}\big)\big] - \big[\widetilde{\tau}_1\big(X^+_{i,0}\big), \widetilde{\tau}_1\big(\widetilde{f}^{r,k}_{i+1,i}\big)\big]\\ &= \big[X^+_{i,1}, \widetilde{f}^{r,k-1}_{i+1,i}\big] - \big[X^+_{i,0}, \widetilde{f}^{r,k}_{i+1,i}\big],\\ \big[X^+_{1,1}, \widetilde{f}^{r,k-1}_{2,1}\big] - \big[X^+_{1,0}, \widetilde{f}^{r,k}_{2,1}\big] &= \big[\widetilde{\tau}_2\big(X^+_{0,1}\big), \widetilde{\tau}_2\big(\widetilde{f}^{r-1,k-1}_{1,-2}\big)\big] - \big[\widetilde{\tau}_2\big(X^+_{0,0}\big), \widetilde{\tau}_2\big(\widetilde{f}^{r-1,k}_{1,-2}\big)\big]\\ &= \big[X^+_{0,1}, \widetilde{f}^{r-1,k-1}_{1,-2}\big] - \big[X^+_{0,0}, \widetilde{f}^{r-1,k}_{1,-2}\big],\\ \big[X^+_{1,1}, \widetilde{f}^{r,k-1}_{2,1}\big] - \big[X^+_{1,0}, \widetilde{f}^{r,k}_{2,1}\big] &= \big[\widetilde{\tau}_3\big(X^+_{n,1}\big), \widetilde{\tau}_3\big(\widetilde{f}^{r,k-1}_{-n,n-1}\big)\big] - \big[\widetilde{\tau}_3\big(X^+_{n,0}\big), \widetilde{\tau}_3\big(\widetilde{f}^{r,k}_{-n,n-1}\big)\big]\\ &= \big[X^+_{n,1}, \widetilde{f}^{r,k-1}_{-n,n-1}\big] - \big[X^+_{n,0}, \widetilde{f}^{r,k}_{-n,n-1}\big],\end{aligned} \tag{3.2}$$

because the right-hand sides are central elements of $D(2n)$. □

Now, we begin the proof of Theorem 3.1. We only consider the case that $r > 0$. When k is even, the images of $a(r,k)$ via Φ are linearly independent. However, in the case that k is odd, we obtain

$$\big[\Phi\big(X^+_{1,1}\big), \Phi\big(\widetilde{f}^{r,k}_{2,1}\big)\big] - \big[\Phi\big(X^+_{1,0}\big), \Phi\big(\widetilde{f}^{r,k}_{2,1}\big)\big] = 0.$$

Thus, we need to show $\big[X^+_{1,1}, \widetilde{f}^{r,2s}_{2,1}\big] - \big[X^+_{1,0}, \widetilde{f}^{r,2s+1}_{2,1}\big] = 0$. The case $s = 0$ is precisely the defining relation. Next, we consider the case $s \ge 1$. By a direct computation, we obtain

$$\begin{aligned}\big[X^+_{n,2}, f^{u,2s-1}_{-n,n-1}\big] &= -\frac{1}{2}\big[X^+_{n,2}, \big[X^+_{n-1,0}, f^{u,2s-1}_{-(n-1),n-1}\big]\big]\\ &= -\frac{1}{2}\big[X^+_{n-1,0}, \big[X^+_{n,2}, f^{u,2s-1}_{-(n-1),n-1}\big]\big] = \big[X^+_{n-1,0}, f^{u,2s+1}_{n,n-1}\big]\end{aligned}$$

and

$$\begin{aligned}\big[X^+_{n,0}, f^{u,2s+1}_{-n,n-1}\big] &= -\frac{1}{2}\big[X^+_{n,0}, \big[X^+_{n-1,2}, f^{u,2s-1}_{-(n-1),n-1}\big]\big] \\ &= -\frac{1}{2}\big[X^+_{n-1,2}, \big[X^+_{n,0}, f^{u,2s-1}_{-(n-1),n-1}\big]\big] = \big[X^+_{n-1,2}, f^{u,2r-1}_{n,n-1}\big].\end{aligned}$$

Thus, we obtain

$$\big[X^+_{n,2}, f^{u,2s-1}_{-n,n-1}\big] - \big[X^+_{n,0}, f^{u,2s+1}_{-n,n-1}\big] = \big[X^+_{n-1,0}, f^{u,2s+1}_{n,n-1}\big] - \big[X^+_{n-1,2}, f^{u,2s-1}_{n,n-1}\big]. \tag{3.3}$$

By (3.1), (3.3) and (3.2), we find the relations

$$\begin{aligned}\big[X^+_{n,2}, f^{u,2s-1}_{-n,n-1}\big] - \big[X^+_{n,0}, f^{u,2s+1}_{-n,n-1}\big] &= 0,\\ \big[X^+_{1,1}, \widetilde{f}^{2r,2s-1}_{2,1}\big] - \big[X^+_{1,0}, \widetilde{f}^{2r+1,k}_{2,1}\big] &= 0.\end{aligned}$$

We complete the proof of Theorem 3.1.

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